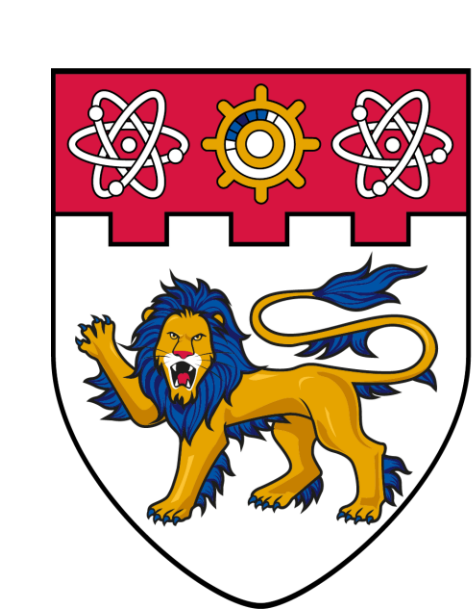




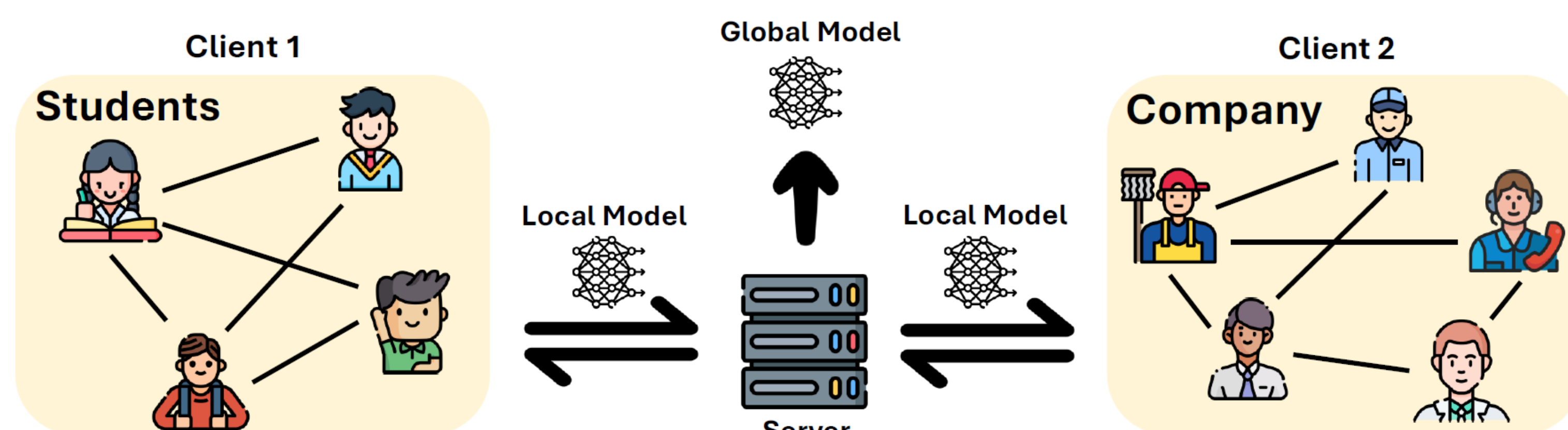
Personalized Subgraph Federated Learning with Differentiable Auxiliary Projections

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Subgraph Federated Learning



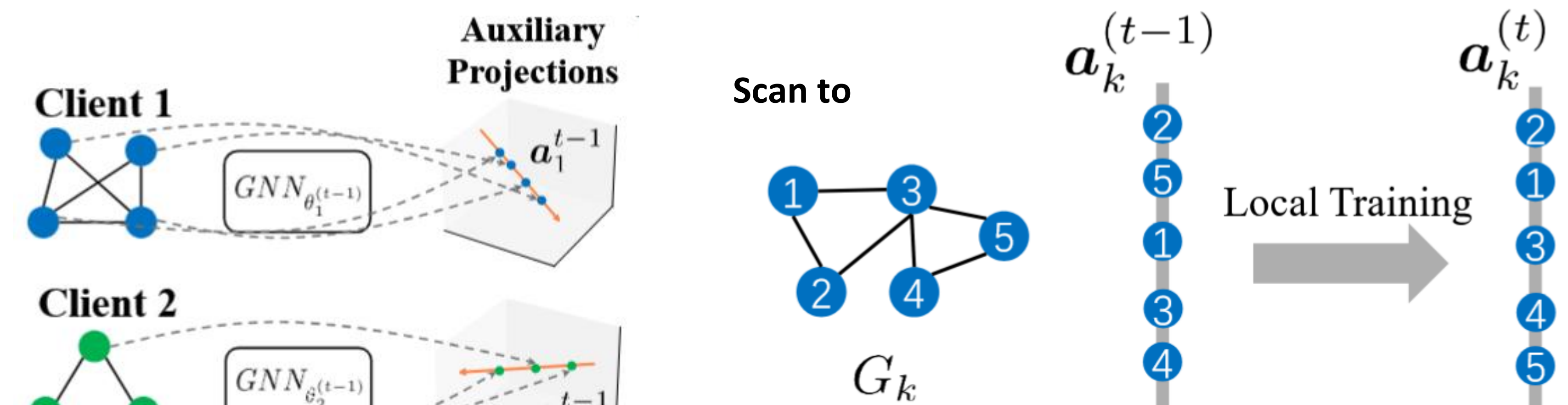
Source of non-IIDness: each client maintains a distinct subgraph of a whole network exhibiting unique interaction patterns, divergent community structures, and heterogeneous feature distributions.

Privacy requirements: exchanging only model parameters between clients and the server, while strictly prohibiting transmission of raw data, gradients, or embeddings.

Challenge:

How to quantify inter-client similarity for personalized subgraph federated learning under severe non-IID heterogeneity and strict privacy constraints?

Auxiliary Projection Vectors (APVs)

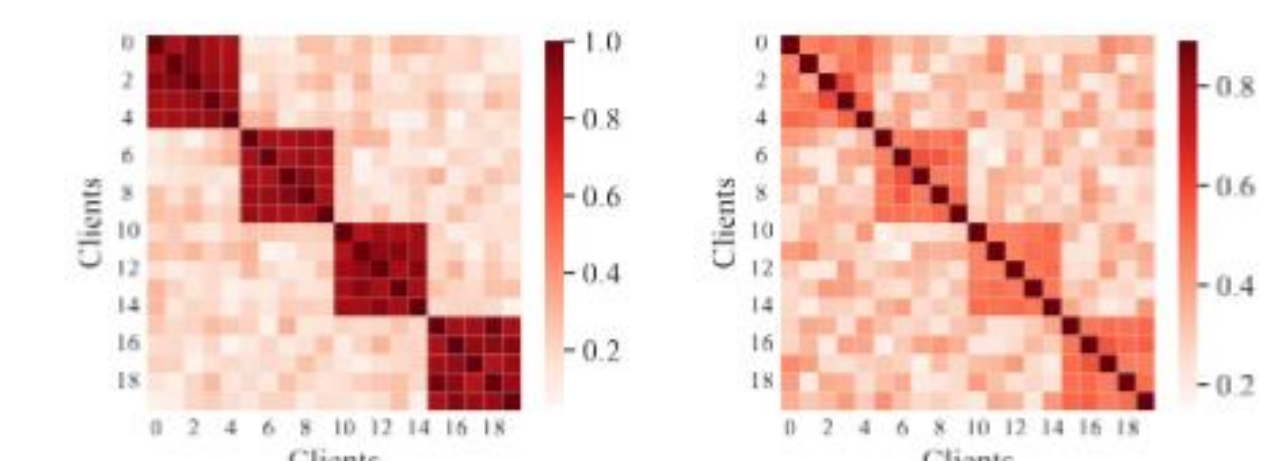
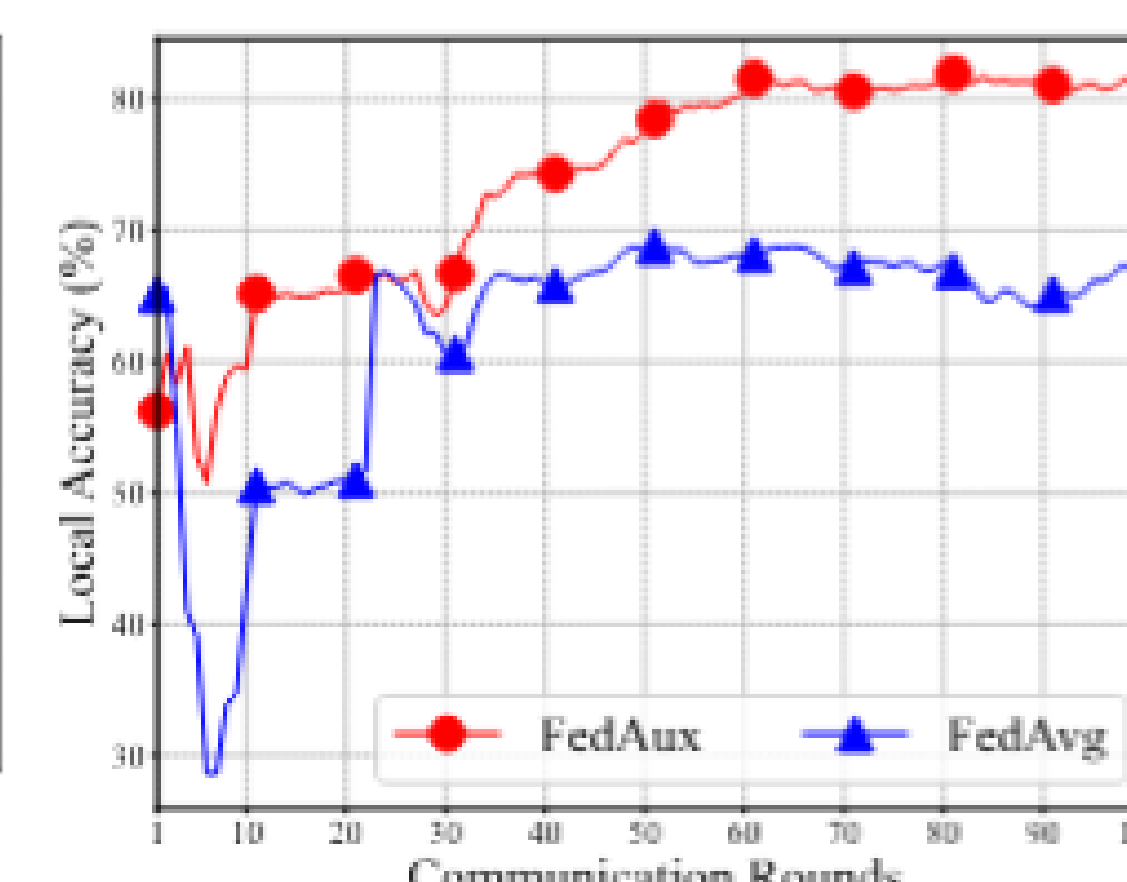
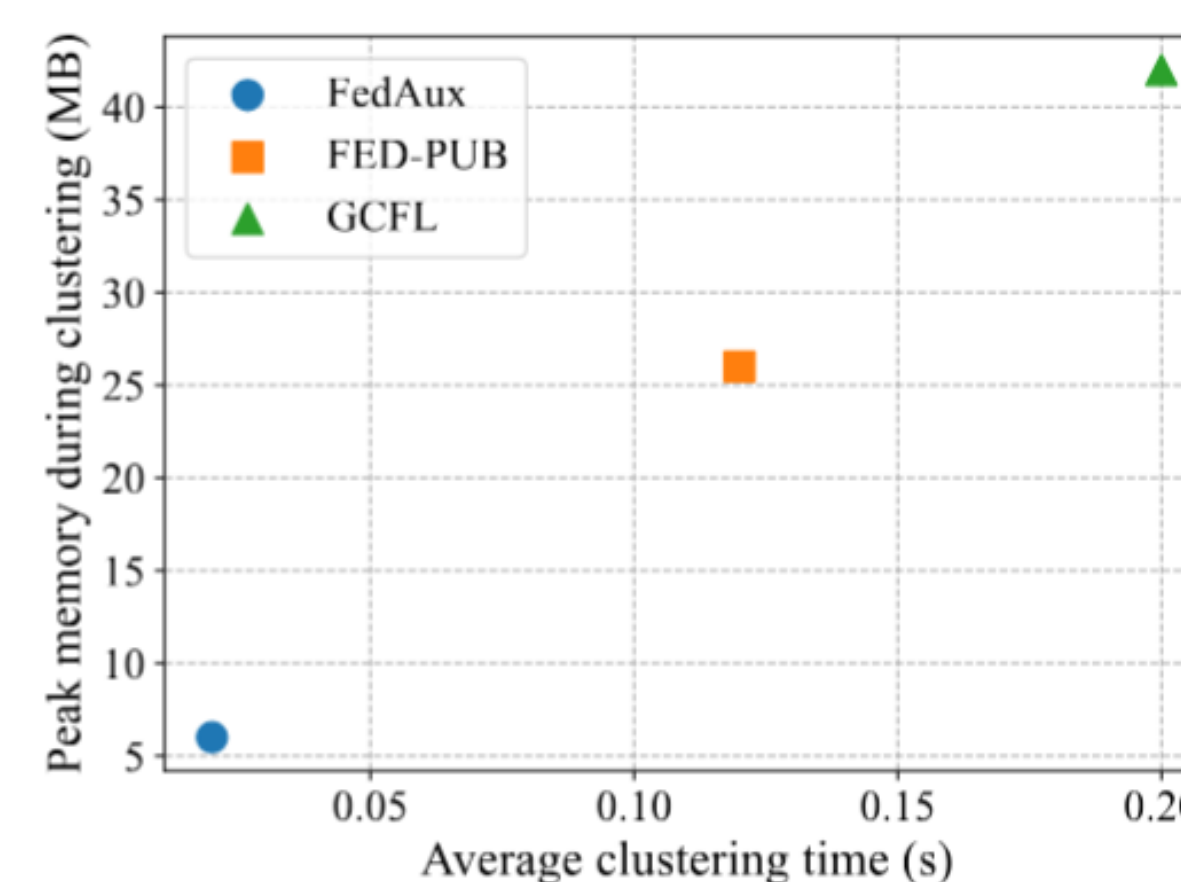


APV client-specific similarity proxies on the server

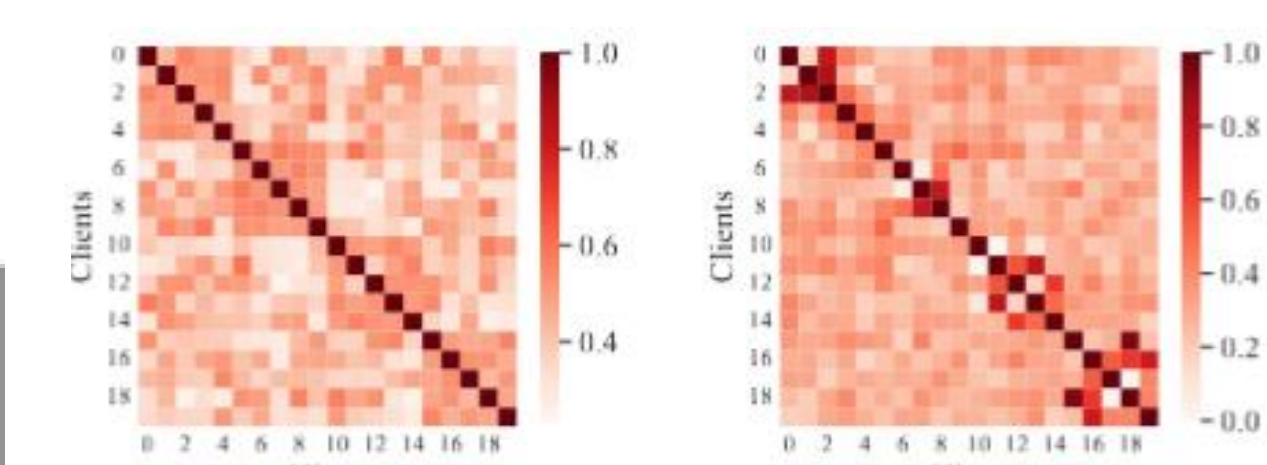
APV serves as a privacy-preserving latent space that retains relative subgraph structure while concealing absolute node positions and raw features.

Experiments

	Amazon-Computer			Amazon-Photo			ogbn-arxiv		
Methods	5 Clients	10 Clients	20 Clients	5 Clients	10 Clients	20 Clients	5 Clients	10 Clients	20 Clients
Local	89.22±0.13	88.91±0.17	89.52±0.20	91.67±0.09	91.80±0.02	90.47±0.15	66.76±0.07	64.92±0.09	65.06±0.05
FedAvg	84.88±1.96	79.54±0.23	74.79±0.24	89.89±0.83	83.15±3.71	81.35±1.04	65.54±0.07	64.44±0.10	63.24±0.13
FedProx	85.25±1.27	83.81±1.09	73.05±1.30	90.38±0.48	80.92±4.64	82.32±0.29	65.21±0.20	64.37±0.18	63.03±0.04
FedPer	89.67±0.34	89.73±0.04	87.86±0.43	91.44±0.37	91.76±0.23	90.59±0.06	66.87±0.05	64.99±0.18	64.66±0.11
GCFL	89.07±0.91	90.03±0.16	89.08±0.25	91.99±0.29	92.06±0.25	90.79±0.17	66.80±0.12	65.09±0.08	65.08±0.04
FedGNN	88.08±0.15	88.18±0.41	83.16±0.13	90.25±0.70	87.12±0.01	81.00±4.48	65.47±0.22	64.21±0.32	63.80±0.05
FedGTA	85.06±0.82	84.27±0.71	79.46±0.28	89.70±0.67	76.53±3.21	82.02±0.78	65.42±0.09	64.22±0.08	63.75±0.18
FedSage+	85.04±0.61	80.50±1.30	70.42±0.85	90.77±0.44	76.81±8.24	80.58±1.15	65.69±0.09	64.52±0.14	63.31±0.20
FED-PUB	90.25±0.07	89.73±0.16	88.20±0.18	93.20±0.15	92.46±0.19	90.59±0.35	67.62±0.11	66.35±0.16	63.90±0.27
FedAux	90.38±0.08	89.92±0.15	88.35±0.96	93.37±0.26	92.30±0.29	90.91±0.60	68.83±0.15	68.50±0.27	65.52±0.10



(a) Embedding Similarity (b) APV-based Similarity



(c) Weight Similarity (d) Functional Similarity

